

Homework 3

Elise LaFreniere

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Part 1. Set up tasks

```
# reading in packages
library(tidyverse)
library(janitor)
library(here)

# reading in kelp data
kelp <- read.csv(
  here("data", "temp-kelp.csv")
)

# reading in personal data finalized
```

Part 2. Problems

Problem 1. Giant kelp fronds

Climate change threatens all ecosystems, including giant kelp (*Macrocystis pyrifera*) forests.

You are interested in the potential relationship between ocean temperature (measured in °C) and giant kelp frond elongation rate (also known as growth rate, measured in cm day⁻¹).

You conduct a field study in which you track giant kelp individuals growing in different temperatures (on average) and measure their elongation rates. Admittedly, this is not a perfect study, but you do the best with what you have!

a. An appropriate test

The appropriate tests to determine the strength of the relationship between ocean temperature and giant kelp frond elongation rate are Pearson's correlation (r) and Spearman's rank correlation (ρ) test. Pearson's correlation (r) evaluates a linear relationship between the continuous variables and assumes normality of the data, whereas Spearman's rank correlation (ρ) assesses the monotonic relationship by ranking the data points. Since biological growth rates like kelp elongation often do not follow a typical linear pattern, Spearman's rank correlation (ρ) may provide a more accurate measure of the relationship strength.

b. Create a visualization

```
# base layer: ggplot
ggplot(kelp, # from data source
       mapping = aes(
         x = temp_c, # x axis temp
         y = kelp_elong # y axis elongation
       )) +
# first layer : data points showing individual observation
geom_point(size = 2,
           alpha = 0.7,
           color = "palegreen4") + # adding custom color
# adding a linear trend line
# adding ribbon to represent confidence interval
geom_smooth(method = "lm",
            color = "sienna",
            fill = "gray",
            alpha = 0.3) +
# renaming axes, adding a title
labs(
  x = "Temperature (°C)",
  y = "Kelp Elongation (cm/day)",
  title = "Giant kelp frond elongation rate in \n response to ocean temperature"
) +
# adding theme
theme_minimal()
```

Giant kelp frond elongation rate in response to ocean temperature

